IP Addressing and Subnetting Network Report

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1. Title Page

Project Title: IP Addressing & Subnetting Implementation
Course: Computer Networks / Network Simulation
Tools Used: Cisco Packet Tracer, Draw.io
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2. Introduction

This project demonstrates the implementation of **IP addressing, subnetting, and routing** in a simulated network using **Cisco Packet Tracer**.
The goal is to create a network for a small organization with **three departments** — HR, Finance, and IT — each with its own subnet.

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3. Objective

- Understand **IP addressing and subnetting** concepts.
- Configure **router interfaces** for inter-subnet communication.
- Configure **switches and PCs** for proper connectivity.
- Test **network connectivity** using ping commands.
- Simulate a **realistic enterprise network** environment.

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4. Network Topology

Description: The network uses a **hierarchical design**:

- **Router (Cisco 2911)** connects to a **Main Core Switch**.
- The **Main Switch** connects to three department switches: HR, Finance, IT.
- Each department switch connects to its PCs.

Topology Diagram:
(Insert screenshot here: network_topology.png)

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5. IP Addressing and Subnetting

Base Network: 192.168.10.0/24

Department	Network Address	Subnet Mask	Gateway	Usable IP Range
HR	192.168.10.0	255.255.255.192	192.168.10.1	192.168.10.1 – 192.168.10.62
Finance	192.168.10.64	255.255.255.192	192.168.10.65	192.168.10.65 – 192.168.10.126
IT	192.168.10.128	255.255.255.192	192.168.10.129	192.168.10.129 – 192.168.10.190

Subnetting Explanation:

- Each department requires up to 62 hosts.
- /26 subnet mask (255.255.255.192) allows 62 usable IPs per subnet.
- Ensures **no IP conflicts** and proper network segmentation.

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6. Device Configuration

Router (Cisco 2911)

Interface	Connected To	IP Address	Subnet Mask
G0/0	Main Switch	192.168.10.1	255.255.255.192
G0/1	Main Switch	192.168.10.65	255.255.255.192
G0/2	Main Switch	192.168.10.129	255.255.255.192

Commands Used:

```
enable
configure terminal
interface gig0/0
ip address 192.168.10.1 255.255.255.192
no shutdown
exit
interface gig0/1
ip address 192.168.10.65 255.255.255.192
no shutdown
exit
interface gig0/2
ip address 192.168.10.129 255.255.255.192
no shutdown
exit
```

Switches (Cisco 2960)

- **Main Switch** connects to router and department switches.
- **Department Switches** connect to PCs.
- **All switch ports enabled** using:

enable

configure terminal

interface range fa0/1-24

no shutdown

exit

PCs

- HR: PC0 → 192.168.10.10, PC1 → 192.168.10.11 (Gateway: 192.168.10.1)
- Finance: PC2 → 192.168.10.70, PC3 → 192.168.10.71 (Gateway: 192.168.10.65)
- IT: PC4 → 192.168.10.130, PC5 → 192.168.10.131 (Gateway: 192.168.10.129)

Configured via Desktop → IP Configuration in Packet Tracer

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7. Testing and Verification

- **Ping tests** were performed to verify connectivity:
  - Within same subnet (HR → HR, Finance → Finance, IT → IT)
  - Across subnets (HR → Finance, Finance → IT, IT → HR)
- **Result:** All pings successful, network fully functional.

Sample Test Command:

PC0> ping 192.168.10.70

Reply from 192.168.10.70: bytes=32 time<1ms TTL=128

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8. Observations

- Router interfaces: up/up
- Switch ports: up
- PCs can communicate within and across subnets.
- No IP conflicts detected.

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9. Conclusion

- Project demonstrates **IP planning, subnetting, and router configuration**.
- Network segmentation ensures **organized departmental structure**.
- Ready for further enhancements:
  - VLAN configuration
  - DHCP for dynamic IP allocation
  - ACLs for security
- This project is suitable for **GitHub portfolio** to showcase networking and subnetting skills.

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10. References

1. Cisco Packet Tracer Simulation Tool
2. Cisco Networking Academy Materials
3. IP Subnetting Calculator: https://www.subnet-calculator.com